Lecture 25

Poincaré representations and little groups

Objectives. *This lecture reduces the classification of positive-energy irreducible unitary representations of the Poincaré group to little groups. It makes the massive and finite-helicity sectors explicit. Translations are diagonalised by momenta. Lorentz transformations move momenta along orbits. The orbit and an irreducible unitary representation of the stabiliser of one of its points determine the representation up to unitary equivalence.*

By the end of this lecture you will be able to:

- Construct the group $\mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ and its action on momenta.
- Classify the momentum orbits and compute the massive and massless little groups.
- State Mackey's classification with its hypotheses, apply it to the $ax+b$ group, and separate the finite-helicity, continuous-spin and zero-momentum sectors.
- Verify the massive spin model and name the steps it omits.

25.1 The Poincaré group as a semidirect product

Spacetime is $\mathbb{R}^4$ with points $x = (x_0, x_1, x_2, x_3)$, where x_0 is time. The *Minkowski pairing* is

$$x \cdot y = x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Lorentz transformations are the linear maps preserving this pairing. The proper orthochronous ones form the group $\mathrm{SO}^+(1,3)$. Adding translations gives the Poincaré group. Quantum symmetries act projectively, and Lecture 13 explains why one then passes to a cover. We prove no lifting theorem. We study unitary representations of the cover built below.

Identify $\mathbb{R}^4$ with the 2×2 Hermitian matrices by

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \qquad \det \Phi(x) = x \cdot x, \qquad \operatorname{tr} \Phi(x) = 2x_0. \tag{25.1}$$

For $g \in \mathrm{SL}(2,\mathbb{C})$ define the map $\Lambda(g)\colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$\Phi(\Lambda(g)x) = g\,\Phi(x)\,g^*.$$

This is the construction of Lecture 24, with Φ in place of X, Λ in place of φ, and $x \cdot y$ in place of $\langle x, y\rangle$.

Lemma 25.1. *The map $\Lambda(g)$ is linear, and its entries are real polynomials in the real and imaginary parts of the entries of g. Moreover $\Lambda(gg') = \Lambda(g)\Lambda(g')$ and $\Lambda(g)x \cdot \Lambda(g)y = x \cdot y$.*

Proof. The matrix $g\Phi(x)g^{*}$ is Hermitian and depends linearly on x. For a Hermitian matrix $h = (h_{ij})$, the vector $\Phi^{-1}(h)$ depends linearly on h. For example, its time coordinate is $x_0 = \tfrac{1}{2}(h_{11} + h_{22})$. Hence $\Lambda(g)$ is linear. Each entry of $g\Phi(x)g^{*}$ is a sum of products of an entry of g, an entry of $\Phi(x)$ and the complex conjugate of an entry of g. So the entries of $\Lambda(g)$ are real polynomials in the real and imaginary parts of the entries of g.

The second claim follows from $gg'\Phi(x)(gg')^{*} = g\big(g'\Phi(x)g'^{*}\big)g^{*}$. For the third, $\det(g\Phi(x)g^{*}) = |\det g|^{2}\det\Phi(x) = x\cdot x$. Polarisation, $2x\cdot y = (x+y)\cdot(x+y) - x\cdot x - y\cdot y$, gives the pairing. ∎

By Lecture 24, Λ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^{+}(1,3)$ with kernel $\{\pm I\}$. The final claim of Proposition 25.3 and the descent statements use this fact. The classification itself does not depend on it.

Definition 25.2. The group $G = \mathbb{R}^{4} \rtimes \mathrm{SL}(2,\mathbb{C})$ is the set $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ with the product

$$(a,g)(a',g') = \big(a + \Lambda(g)a',\; gg'\big). \tag{25.2}$$

Proposition 25.3. *G is a Lie group of dimension 10. Its identity is $(0,I)$ and $(a,g)^{-1} = \big(-\Lambda(g)^{-1}a,\; g^{-1}\big)$. The translations $A = \{(a,I)\} \cong \mathbb{R}^{4}$ form a closed normal subgroup, and*

$$(0,g)(a,I)(0,g)^{-1} = \big(\Lambda(g)a,\; I\big). \tag{25.3}$$

The map $(a,g) \mapsto (a,\Lambda(g))$ is a homomorphism onto $\mathbb{R}^{4} \rtimes \mathrm{SO}^{+}(1,3)$ with kernel $\{(0,\pm I)\}$.

Proof. Associativity is the main point. Both bracketings of a triple product have second entry $gg'g''$. Their first entries are

$$a + \Lambda(g)a' + \Lambda(gg')a'' \qquad \text{and} \qquad a + \Lambda(g)\big(a' + \Lambda(g')a''\big).$$

These agree because $\Lambda(g)$ is linear and Λ is multiplicative, by Lemma 25.1. The stated inverse is an inverse on both sides, since

$$(a,g)\big(-\Lambda(g)^{-1}a,\, g^{-1}\big) = (a-a,\; I), \qquad \big(-\Lambda(g)^{-1}a,\, g^{-1}\big)(a,g) = \big(-\Lambda(g)^{-1}a + \Lambda(g)^{-1}a,\; I\big).$$

The manifold $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ has dimension $4 + 6$. Multiplication is polynomial in real coordinates by Lemma 25.1. So is inversion, since g^{-1} is the adjugate of g. Hence G is a Lie group. The projection $(a,g) \mapsto g$ is a continuous homomorphism with kernel A, so A is closed and normal. Next, $(0,g)(a,I) = (\Lambda(g)a, g)$, and multiplying on the right by $(0,g^{-1})$ gives (25.3). The final map respects (25.2), because only $\Lambda(g)$ acts on translations. Its image and kernel come from those of Λ. ∎

From now on, the Poincaré group means G. The central element $z = (0,-I)$ maps to the identity.

25.2 Momentum characters and mass shells

Let U be a strongly continuous unitary representation of G on a Hilbert space. On A it gives commuting unitary operators. Their building blocks are characters. Exercise 25.2 proves the next lemma.

> **Lemma 25.4.** *Every continuous homomorphism $\mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.*

The sign follows the generator convention of Lecture 15. For the one-dimensional representation χ_p and the translation generator $X = e_0$, we have $\chi_p(te_0) = e^{-itp_0}$. Thus $A_X = p_0$ in that convention, and p_0 is the energy.

To describe all these operators together we use a projection-valued measure. We recall this object from *Linear Algebra for Physics*, Lecture 14. There the spectral theorem attaches to a self-adjoint operator a projection $P(\Omega)$ for each Borel subset Ω of its spectrum. The Borel sets are obtained from the open sets by countable unions, intersections and complements. For multiplication by x on $L^2[0,1]$, the projection $P(\Omega)$ is multiplication by the indicator function of Ω, and each single point has $P(\{\lambda\}) = 0$.

The same definition works on $\mathbb{R}^4$. A *projection-valued measure* E on $\mathbb{R}^4$ assigns an orthogonal projection $E(S)$ to every Borel set $S \subseteq \mathbb{R}^4$, with $E(\mathbb{R}^4) = I$ and countable additivity in the strong operator topology for disjoint sets. We say that E is *concentrated* on a Borel set S when $E(S) = I$.

We cite the *Stone–Naimark–Ambrose–Godement (SNAG) theorem* for vector groups; see Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2. For a strongly continuous unitary representation of $\mathbb{R}^4$, it gives a unique projection-valued measure E on $\mathbb{R}^4$ with

$$U(a, I) = \int_{\mathbb{R}^4} \chi_p(a)\, dE(p).$$

Here $E(S)$ is the projection onto the part of the space whose momentum lies in S. For a vector v, the map $S \mapsto \langle v, E(S)v\rangle$ is a positive Borel measure of total mass $\|v\|^2$. For a unit vector v, it is the probability distribution of the momentum in the state v, as in the Born rule of Lecture 14 of that course. Unlike the measures of Lecture 17, it may vanish on open sets. The operator integral has the following meaning: for every vector v,

$$\langle v, U(a, I)v\rangle = \int_{\mathbb{R}^4} \chi_p(a)\, d\langle v, E(p)v\rangle.$$

By polarisation, these scalar identities determine the bounded operator $U(a, I)$.

Example 25.5 (A multiplication model)**.** On $L^2(\mathbb{R}^4, d^4p)$ let $(U(a)\psi)(p) = e^{-ip\cdot a}\psi(p)$, and let $E(S)$ multiply ψ by the indicator function of S. Then

$$\langle \psi, E(S)\psi\rangle = \int_S |\psi(p)|^2\, d^4p, \qquad \langle \psi, U(a)\psi\rangle = \int_{\mathbb{R}^4} e^{-ip\cdot a}|\psi(p)|^2\, d^4p.$$

So this E is the measure that the SNAG theorem attaches to U. A single point has Lebesgue measure zero, so $E(\{p\}) = 0$ for every p. This representation therefore has no momentum eigenvector in the Hilbert space, although every vector has a momentum distribution.

The *support* of E is the complement of the union of open sets O with $E(O) = 0$. Since $\mathbb{R}^4$ has a countable basis, this union is a countable union of such null sets. Thus the support is the smallest closed set F with $E(F) = I$. We call it the *translation spectrum*.

We use the spectral calculus for projection-valued measures as a cited analytic input (Hall, *Quantum Theory for Mathematicians*, §10.1). It defines the self-adjoint operator

$$P_0 = \int_{\mathbb{R}^4} p_0 \, dE(p), \qquad \mathrm{Dom}(P_0) = \left\{ v : \int_{\mathbb{R}^4} p_0^2 \, d\langle v, E(p)v\rangle < \infty \right\}.$$

By Stone's theorem with the convention of Lecture 15, this is the generator of $t \mapsto U(te_0, I)$. Positive energy means $P_0 \geq 0$. By the spectral calculus this is equivalent to concentration of E on $\{p_0 \geq 0\}$. Momentum eigenvectors need not exist in the Hilbert space.

Still, a formal computation shows the mechanism. If $U(a, I)\psi = \chi_p(a)\psi$ for all a, then (25.3) gives

$$U(a, I)U(0, g)\psi = U(0, g)\, U(\Lambda(g)^{-1}a, I)\psi = \chi_{\Lambda(g)p}(a)\, U(0, g)\psi. \tag{25.4}$$

Here $p \cdot \Lambda(g)^{-1}a = \Lambda(g)p \cdot a$ by the pairing invariance in Lemma 25.1. In particular, if g fixes p, then $U(0, g)\psi$ again has momentum p. So the stabiliser of p acts on the formal eigenvectors of momentum p. Moving this action around the orbit of p gives the whole space. This is the idea of induction, and it is why stabilisers appear in the classification below.

The same calculation with E, together with the uniqueness of E in the SNAG theorem, gives

$$U(0, g)E(S)U(0, g)^{-1} = E(\Lambda(g)S).$$

Hence the translation spectrum is stable under the following action.

Definition 25.6. The *mass-shell action* of $\mathrm{SL}(2, \mathbb{C})$ on momentum space $\mathbb{R}^4$ is $g \cdot p = \Lambda(g)p$. For $m, \kappa > 0$ set

$$X_m^{\pm} = \{p \cdot p = m^2,\ \pm p_0 > 0\}, \qquad C^{\pm} = \{p \cdot p = 0,\ \pm p_0 > 0\}, \qquad Y_\kappa = \{p \cdot p = -\kappa^2\}.$$

Theorem 25.7. *The orbits of the mass-shell action are exactly the sets $X_m^{\pm}$, $C^{\pm}$, Y_κ and $\{0\}$.*

Proof. Invariants. The number $p \cdot p$ is invariant, because $\Lambda(g)$ preserves the pairing by Lemma 25.1. Write $\vec{p} = (p_1, p_2, p_3)$, with Euclidean norm $|\vec{p}|$. Suppose $p \cdot p \geq 0$ and $p \neq 0$. Then $p_0^2 \geq |\vec{p}|^2$, so $p_0 \neq 0$. The eigenvalues $\mu_1 \geq \mu_2$ of $\Phi(p)$ satisfy

$$\mu_1\mu_2 = p \cdot p \geq 0, \qquad \mu_1 + \mu_2 = 2p_0.$$

So $p_0 > 0$ exactly when $\Phi(p)$ is positive semidefinite and nonzero. The map $\Phi \mapsto g\Phi g^*$ preserves both properties, because $v^* g\Phi g^* v = (g^* v)^* \Phi(g^* v)$ and g is invertible. Hence each listed set is a union of orbits. The listed sets partition $\mathbb{R}^4$.

Transitivity. Write $\Phi(p) = u\ \mathrm{diag}(\mu_1, \mu_2)\, u^*$ with $u \in \mathrm{U}(2)$. A scalar factor $e^{i\phi}$ with $e^{2i\phi} \det u = 1$ puts u in $\mathrm{SU}(2)$ and changes nothing. Choose the base points

$$k_m = (m, 0, 0, 0), \qquad k = (1, 0, 0, 1), \qquad \ell_\kappa = (0, 0, 0, \kappa).$$

Their Φ-images are $\mathrm{diag}(m, m)$, $\mathrm{diag}(2, 0)$ and $\mathrm{diag}(\kappa, -\kappa)$. For $s > 0$ let $d_s = \mathrm{diag}(s, s^{-1})$. Then

$$d_s\ \mathrm{diag}(\alpha, \beta)\, d_s^* = \mathrm{diag}(s^2\alpha,\ s^{-2}\beta). \tag{25.5}$$

On X_m^+ we have $\mu_1, \mu_2 > 0$ and $\mu_1\mu_2 = m^2$. Take $s^2 = \mu_1/m$. Then $s^{-2}m = \mu_2$, so $p = \Lambda(ud_s)k_m$. On C^+ we have $\mu_1 = 2p_0$ and $\mu_2 = 0$; take $s^2 = \mu_1/2$. On Y_κ we have $\mu_1 > 0 > \mu_2$ and $\mu_1\mu_2 = -\kappa^2$; take $s^2 = \mu_1/\kappa$. Since $\Lambda(g)$ is linear, X_m^- and C^- are the orbits of $-k_m$ and $-k$. Finally 0 is fixed. ∎

Figure 25.1 draws these orbits with one spatial coordinate suppressed. The base points k_m, k and ℓ_κ all have $p_1 = p_2 = 0$, and their stabilisers are computed in the next section.

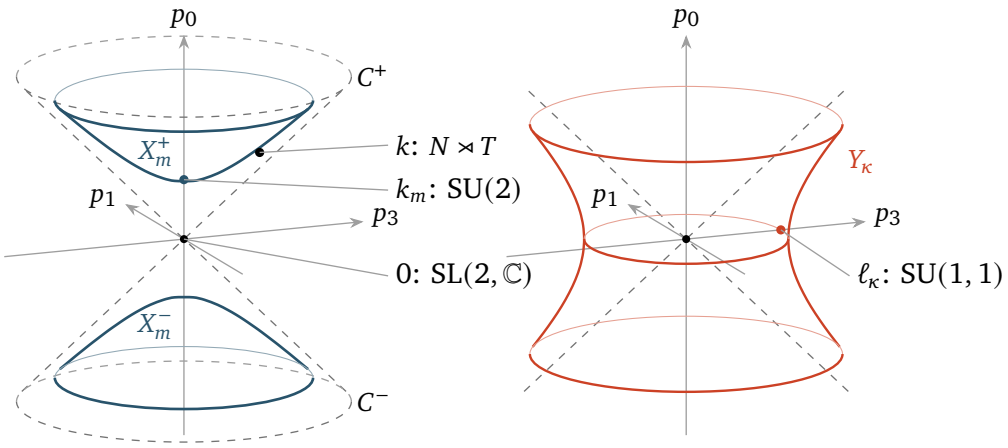


Figure 25.1: The orbits of Theorem 25.7 in the slice $p_2 = 0$. Each drawn surface is a slice of a three-dimensional orbit, since the coordinate p_2 is suppressed. In the panel with the two bowls, the bowls are $X_m^\pm$ and the dashed cone is $C^\pm$ with its vertex 0 removed. In the other panel, the connected waisted surface is Y_κ, around the same cone. Each marked base point carries the name of its stabiliser. Positive energy keeps the upper bowl, the upper cone and the origin. It excludes all of Y_κ, including the points of Y_κ with $p_0 > 0$.

By (25.5), $\Lambda(d_s)$ multiplies $p_0 + p_3$ by s^2 and $p_0 - p_3$ by s^{-2}. It is a boost along the x_3-axis.

A subset of $\mathbb{R}^4$ is *locally closed* when it is the intersection of a closed set and an open set. Each orbit is locally closed. For example, C^+ is the closed set $\{p \cdot p = 0\}$ intersected with the open set $\{p_0 > 0\}$.

The invariant Borel sets $\{p \cdot p < r\}$ with r rational, together with $\{p \cdot p \geq 0,\ p_0 > 0\}$, $\{p \cdot p \geq 0,\ p_0 < 0\}$ and $\{0\}$, separate the orbits. Local closedness and this separation property form the regularity used below.

These orbits lie in the dual of the translation subgroup. They are not the coadjoint orbits of Lecture 19, which lie in the dual of the whole Lie algebra. The two kinds are related. Restricting a functional to the translations maps a coadjoint orbit onto one orbit in the dual of the translations, because translations act trivially on that dual. With $\chi_\ell = e^{i\ell}$ in Lecture 19 and $\chi_p = e^{-ip}$ here, the restriction of ℓ is the functional $-p$. For the $ax + b$ group, this restriction sends the half-planes of Exercise 19.1 onto the frequency half-lines of Lecture 17.

25.3 Little groups and Mackey's classification

Definition 25.8. The *little group* of $p \in \mathbb{R}^4$ is its stabiliser $H_p = \{g \in \mathrm{SL}(2,\mathbb{C}) : \Lambda(g)p = p\}$.

For $b \in \mathbb{C}$ and $|\lambda| = 1$ write

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \bar\lambda \end{pmatrix}.$$

Proposition 25.9. *(a)* $H_{k_m} = \mathrm{SU}(2)$.
(b) $H_k = \{n(b)r(\lambda) : b \in \mathbb{C},\ |\lambda| = 1\}$, *and* $r(\lambda)n(b)r(\lambda)^{-1} = n(\lambda^2 b)$.
(c) $\Lambda(r(e^{-i\theta/2}))$ *rotates the* (x_1, x_2)*-plane by the angle* $+\theta$ *and fixes* x_0 *and* x_3.
(d) $H_{\ell_\kappa} = \{g : g\sigma_3 g^* = \sigma_3\} = \mathrm{SU}(1,1)$, *where* $\sigma_3 = \mathrm{diag}(1,-1)$. *Also* $H_0 = \mathrm{SL}(2,\mathbb{C})$.

Proof. (a) The condition is $g(mI)g^* = mI$, that is, $gg^* = I$.

(b) Since $\Phi(k) = 2e_1e_1^*$, the condition is $vv^* = e_1e_1^*$ for $v = ge_1$. In entries,

$$\begin{pmatrix} |v_1|^2 & v_1\bar v_2 \\ v_2\bar v_1 & |v_2|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $v_2 = 0$ and $v_1 = \lambda$ with $|\lambda| = 1$. Since $\det g = 1$,

$$g = \begin{pmatrix} \lambda & w \\ 0 & \bar\lambda \end{pmatrix} = n(\lambda w)\, r(\lambda).$$

Every such g fixes k. The conjugation formula is a product of three matrices:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar\lambda \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \bar\lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 & \lambda^2 b \\ 0 & 1 \end{pmatrix}.$$

(c) Conjugation by $r(\lambda)$ fixes the diagonal of $\Phi(p)$. It multiplies the entry $p_1 + ip_2$ by $\bar\lambda^2$, which is $e^{i\theta}$ for $\lambda = e^{-i\theta/2}$.

(d) This restates the definition, since $\Phi(\ell_\kappa) = \kappa\sigma_3$. ∎

The group in part (d) is the group $\mathrm{SU}(1,1)$ of Lecture 23. It is also a copy of the real subgroup of Lecture 24. Every $g \in \mathrm{SL}(2,\mathbb{C})$ satisfies $g\sigma_2 g^T = \sigma_2$, so $g\sigma_2 g^* = \sigma_2$ exactly when g is real. Thus $\mathrm{SL}(2,\mathbb{R})$ is the stabiliser of the spacelike vector e_2. The Cayley matrix γ of Lecture 23 satisfies $\gamma\gamma^* = 2I$, so a scalar multiple u of γ lies in $\mathrm{SU}(2)$. It also satisfies $\gamma\sigma_2\gamma^{-1} = -\sigma_3$, so $\Lambda(u)e_2 = -e_3$. Hence $g \mapsto \gamma g \gamma^{-1}$ carries $\mathrm{SL}(2,\mathbb{R})$ onto the stabiliser of e_3, which is H_{ℓ_κ}. This does not make the spacelike orbit relevant to positive energy. It identifies the little-group data of Y_κ with the classification of Lecture 23.

Write $N = \{n(b)\} \cong \mathbb{C}$ and $T = \{r(\lambda)\} \cong \mathrm{U}(1)$. By the factorisation in Proposition 25.9(b), $H_k = N \rtimes T$. The map $n(b)r(\lambda) \mapsto (b, \lambda^2)$ is two-to-one onto the group of orientation-preserving Euclidean motions of the plane. So we call N the *little-group translations*. The maps $\Lambda(n(b))$ are Lorentz transformations fixing k, not spacetime translations. For $b = 1$ this map is the null rotation of Exercise 24.2.

We have $r(e^{-i\theta/2}) = \exp(\theta X_3)$ with $X_3 = -i\sigma_3/2$. By the rotation formula in Proposition 25.9(c), this generates positive-angle rotations about x_3, as in Lecture 24.

The next theorem uses unitary induction from Lecture 17.

Theorem 25.10 (Mackey, cited). *Let $G = \mathbb{R}^n \rtimes H$, where H is a second-countable Lie group acting linearly and continuously on $\mathbb{R}^n$. For $p \in (\mathbb{R}^n)^*$ put $\chi_p(a) = e^{-ip(a)}$. Let H act on these momenta so that $\chi_{h\cdot p}(a) = \chi_p(h^{-1}\cdot a)$. Assume the action is* regular*: orbits are locally closed, and countably many invariant Borel sets separate them. Let H_p be the stabiliser of p. For a strongly continuous unitary representation U of G, let E be the projection-valued measure on $(\mathbb{R}^n)^*$ that the SNAG theorem attaches to the translations $a \mapsto U(a, e)$.*

(a) If U is irreducible, then E is concentrated on one orbit Ω.

(b) Let $p \in \Omega$ and let σ be a strongly continuous irreducible unitary representation of H_p. Put $(\chi_p \otimes \sigma)(a, h) = \chi_p(a)\sigma(h)$. Then $\operatorname{Ind}_{\mathbb{R}^n \rtimes H_p}^{G}(\chi_p \otimes \sigma)$ is irreducible, and its measure E is concentrated on Ω.

(c) Every strongly continuous irreducible unitary representation whose measure E is concentrated on Ω is unitarily equivalent to one of the induced representations in (b). For a fixed $p \in \Omega$, two of these induced representations are unitarily equivalent exactly when their representations σ are unitarily equivalent.

This is Mackey's method for semidirect products, as presented in A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. For our G it is Wigner's classification of 1939. For our spacetime group, the Minkowski pairing identifies the functionals $p \in (\mathbb{R}^4)^*$ of the theorem with the vectors p used above. The remarks after the orbit theorem, Theorem 25.7, verify regularity. The dual action is $g \cdot p = \Lambda(g)p$ by (25.4).

Before the ten-dimensional group we run the theorem on two small groups from earlier lectures.

Example 25.11 (The $ax+b$ and Heisenberg groups). The $ax+b$ group of Lecture 17 is $\mathbb{R} \rtimes \mathbb{R}_{>0}$, where $a > 0$ acts on the translations n_c by scaling. We keep the frequency ξ of Lecture 17, with $\chi_\xi(n_c) = e^{i\xi c}$, so the momentum of the theorem is $p = -\xi$. The element a sends ξ to ξ/a. The orbits are $\xi > 0$, $\xi < 0$ and $\{0\}$. These three sets are locally closed and separate the orbits, so the action is regular.

The orbits $\xi > 0$ and $\xi < 0$ have trivial stabiliser. By part (b), each gives one irreducible representation, induced from χ_ξ on N. These are Π_+ and Π_- of Lecture 17, so both are irreducible. The orbit $\{0\}$ has stabiliser $\mathbb{R}_{>0}$. The irreducible unitary representations of $\mathbb{R}_{>0}$ are the characters $a \mapsto a^{is}$, with $s \in \mathbb{R}$. They give the one-dimensional representations $(a, b) \mapsto a^{is}$. By parts (a) and (c), this is the whole list of irreducible unitary representations. In particular, the representation Π_0 of Lecture 17 is not irreducible. Translations act trivially in it, so its measure is concentrated on $\{0\}$, and it is infinite-dimensional.

The Heisenberg group of Lecture 18 is also a semidirect product. The subgroup $A = \{(0, b, c)\}$ is a closed normal vector group, the subgroup $\{(a, 0, 0)\}$ is a copy of $\mathbb{R}$, and

$$(a, 0, 0)(0, b, c)(-a, 0, 0) = (0, b, c + ab).$$

Write a character of A as $e^{i(\mu b + \lambda c)}$. The element $(a, 0, 0)$ sends (μ, λ) to $(\mu - \lambda a, \lambda)$. For $\lambda \neq 0$, the line with this value of λ is one orbit with trivial stabiliser. So exactly one irreducible unitary representation has central character λ. By Exercise 18.5 it is the Schrödinger representation Π_λ. This is the Stone–von Neumann theorem for irreducible representations. For $\lambda = 0$, the orbits are single points $(\mu, 0)$ with stabiliser $\mathbb{R}$. The characters of this stabiliser give the characters $\chi_{\alpha,\beta}$ of Lecture 18. All these orbits are closed. Rational levels of λ, and of μ on the line $\lambda = 0$, separate them. So this action is regular too.

Corollary 25.12. *(a) For $m > 0$, up to unitary equivalence, the strongly continuous irreducible unitary representations of G with E concentrated on X_m^+ are labelled by $n \geq 0$. The label is the $\mathrm{SU}(2)$-representation V_n, of dimension $n+1$ and spin $n/2$.*

(b) Up to unitary equivalence, those with E concentrated on C^+ are labelled by the equivalence classes of strongly continuous irreducible unitary representations of $H_k = N \rtimes T$.

Proof. Combine Mackey's theorem, Theorem 25.10, with the little groups computed in Proposition 25.9. In the massive case, strongly continuous irreducible unitary representations of the compact group $\mathrm{SU}(2)$ are finite-dimensional by Lecture 10. Hence they are the V_n of Lecture 8. ∎

25.4 Massless sectors: helicity and continuous spin

Proposition 25.13. *Let σ be a finite-dimensional continuous unitary representation of H_k on V. Then $\sigma(n(b)) = I$ for all b. If σ is irreducible, then $\sigma(n(b)r(\lambda)) = \lambda^\nu$ for a unique $\nu \in \mathbb{Z}$, and*

$$\sigma(\exp \theta X_3) = e^{-i\theta\nu/2}.$$

Proof. The operators $\sigma(n(b))$ are unitary and commute. So V is the direct sum of their joint eigenspaces. On each, $b \mapsto \sigma(n(b))$ is a continuous character of $\mathbb{C} \cong \mathbb{R}^2$. As in the character lemma, Lemma 25.4, it has the form $b \mapsto e^{-i\,\mathrm{Re}(\bar\zeta b)}$ for a unique $\zeta \in \mathbb{C}$. Let V_ζ be the joint eigenspace for ζ. Let S be the finite set of ζ with $V_\zeta \neq 0$.

This is the key step. By the conjugation formula in Proposition 25.9(b), $n(b)r(\lambda) = r(\lambda)n(\bar\lambda^2 b)$. Take $v \in V_\zeta$. Then

$$\sigma(n(b))\sigma(r(\lambda))v = \sigma(r(\lambda))\sigma(n(\bar\lambda^2 b))v = e^{-i\,\mathrm{Re}(\overline{\lambda^2\zeta}\, b)}\,\sigma(r(\lambda))v.$$

So $\sigma(r(\lambda))V_\zeta = V_{\lambda^2\zeta}$. Every unit complex number is some λ^2. If some $\zeta \neq 0$ lay in S, the whole circle $|\zeta'| = |\zeta|$ would lie in S. This contradicts finiteness. So $S = \{0\}$, and N acts trivially.

Then σ factors through $n(b)r(\lambda) \mapsto \lambda$. This is a homomorphism onto $\mathrm{U}(1)$, since $n(b)r(\lambda)n(b')r(\lambda') = n(b + \lambda^2 b')r(\lambda\lambda')$. The operators $\sigma(r(\lambda))$ commute. If σ is irreducible, a common eigenline is invariant, so $\dim V = 1$. By the one-variable character argument of Exercise 25.2(a), $\theta \mapsto \sigma(r(e^{i\theta}))$ equals $e^{i\nu\theta}$ for some real ν. Periodicity in θ forces $\nu \in \mathbb{Z}$. Finally $\exp \theta X_3 = r(e^{-i\theta/2})$ maps to $e^{-i\nu\theta/2}$. ∎

Compare with the generator convention $\Pi(\exp \theta X) = \exp(-i\theta A_X)$ of Lecture 15. The self-adjoint generator of rotations about the momentum direction is $A_{X_3} = \nu/2$. This number is the *helicity*.

The *finite-helicity* massless representations are the representations induced from $\chi_k \otimes \sigma$, where σ is a finite-dimensional continuous irreducible unitary representation of H_k. By Proposition 25.13, these are the representations $\mathrm{Ind}(\chi_k \otimes \sigma_\nu)$ with $\nu \in \mathbb{Z}$ and $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$. Each has the single helicity $\nu/2$. In these representations the little-group translations act trivially.

Remark 25.14. Mackey's theorem, Theorem 25.10, also applies to $H_k = N \rtimes T$. The T-orbits on the characters of N are $\{0\}$ and the circles $|\zeta| = \rho > 0$. These orbits are closed. The sets $\{|\zeta| < r\}$ with $r \in \mathbb{Q}$, together with $\{0\}$, separate them. Hence the action is regular.

The orbit $\{0\}$ gives the σ_ν. A circle has stabiliser $\{r(\pm 1)\}$ with two characters. These give infinite-dimensional irreducible representations of H_k that are nontrivial on N (Exercise 25.7). Inducing them to G gives the *continuous-spin* representations. Their spectral measure is concentrated on C^+, so their energy is positive. Positive energy alone does not exclude them. They form a sector separate from finite helicity.

Now suppose a nonzero representation has E concentrated on C^+. Its closed support meets C^+ and is Lorentz-invariant by covariance of E. Transitivity then puts all of C^+ in the support. Closedness adds its limit point 0. So concentration on an orbit differs from equality of closed supports.

The orbit $\{0\}$ is also separate. Translations act trivially there. Any strongly continuous irreducible unitary representation of $\mathrm{SL}(2,\mathbb{C})$ gives a representation of this zero-momentum sector by composition with the projection $G \to \mathrm{SL}(2,\mathbb{C})$. The trivial one is the usual vacuum choice. Positive energy alone does not select it.

Positive energy excludes X_m^- and C^-, where $p_0 < 0$. It also excludes Y_κ, because this orbit contains points with $p_0 < 0$ and the group moves every point of it to such points. In detail, the set $S = Y_\kappa \cap \{p_0 < 0\}$ is nonempty and relatively open. By transitivity, the Lorentz translates of S cover Y_κ. By second countability, countably many of these translates already cover Y_κ. Positive energy gives $E(S) = 0$, since $S \subset \{p_0 < 0\}$. Then covariance of E makes every translate of S an E-null set. Disjointising this countable cover and using countable additivity would give $E(Y_\kappa) = 0$. This contradicts concentration of E on that orbit in a nonzero representation.

The table collects the sectors. Here $\rho > 0$ is the radius of a T-orbit on the characters of N, and the sign is the value at $r(-1)$ of a character of its stabiliser $\{r(\pm 1)\}$.

Orbit	Base point	Stabiliser	Labels	Energy ≥ 0
X_m^+	k_m	$\mathrm{SU}(2)$	V_n, $n \geq 0$	yes
C^+, N trivial	k	$N \rtimes T$	σ_ν, $\nu \in \mathbb{Z}$	yes
C^+, N nontrivial	k	$N \rtimes T$	$\rho > 0$ and a sign	yes
$\{0\}$	0	$\mathrm{SL}(2,\mathbb{C})$	not computed here	yes
Y_κ	ℓ_κ	$\mathrm{SU}(1,1)$	the list of Lecture 23	no
X_m^-, C^-	$-k_m$, $-k$	as for X_m^+, C^+	as for X_m^+, C^+	no

In physics language, X_m^+ with $n = 0$, 1, 2 describes massive particles such as the pion, the electron and the W boson, while C^+ with $\nu = \pm 2$ describes the photon and $\nu = \pm 4$ the graviton, and no known particle has continuous spin.

25.5 The massive induced model

Fix $m > 0$ and $n \geq 0$. Let D_n be the irreducible unitary representation of $\mathrm{SU}(2)$ on V_n. Put $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$. Define a measure μ_m on X_m^+ by

$$\int_{X_m^+} f \, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p}) \, \frac{d^3\vec{p}}{2\,\omega(\vec{p})}.$$

The factor $1/\omega$ makes this measure invariant. For a boost along x_3 this is visible at once. Put $p_\perp = (p_1, p_2)$ and write $p_3 = \sqrt{m^2 + |p_\perp|^2}\,\sinh\eta$. Then $\omega = \sqrt{m^2 + |p_\perp|^2}\,\cosh\eta$ and $dp_3/\omega = d\eta$. A boost along x_3 fixes $p_\perp$ and shifts the rapidity η. Exercise 25.4 proves invariance under all of $\mathrm{SL}(2,\mathbb{C})$.

For $p \in X_m^+$ the matrix $\Phi(p)$ is positive definite. Let $h_p = (\Phi(p)/m)^{1/2}$ be its positive square root. Then $\det h_p = 1$ and $h_p (mI) h_p^* = \Phi(p)$, so $\Lambda(h_p) k_m = p$. The *Wigner rotation* is

$$W(g, p) = h_p^{-1}\, g\, h_{\Lambda(g)^{-1}p}.$$

Put $q = \Lambda(g)^{-1}p$. Read from the right, the three factors move the base point k_m to q, then q to p, then p back to k_m. Figure 25.2 shows this loop for the exact data of Exercise 25.5(e).

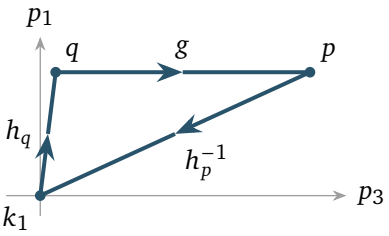


Figure 25.2: The loop of the Wigner rotation for $m = 1$ and $g = \mathrm{diag}(2, \tfrac{1}{2})$. The slice $p_2 = 0$ of X_1^+ is drawn as a graph over the (p_3, p_1)-plane, with $q = (\tfrac{713}{128}, \tfrac{87}{16}, 0, \tfrac{87}{128})$ and $p = (\tfrac{6713}{512}, \tfrac{87}{16}, 0, \tfrac{6087}{512})$. The boost h_q moves k_1 to q, the boost g moves q to p, and the boost h_p^{-1} moves p back to k_1. The product fixes k_1, so it lies in $\mathrm{SU}(2)$. It acts on the spin fibre V_n, which the drawing of the base does not show.

The model space is $\mathcal{H}_{m,n} = L^2(X_m^+, \mu_m; V_n)$. The operators are

$$(U(a,g)\psi)(p) = e^{-ip\cdot a}\, D_n\big(W(g,p)\big)\, \psi(\Lambda(g)^{-1}p). \tag{25.6}$$

Proposition 25.15. *(a)* $W(g,p) \in \mathrm{SU}(2)$.
(b) $W(gg', p) = W(g,p)\,W(g', \Lambda(g)^{-1}p)$.
(c) $U((a,g)(a',g')) = U(a,g)\,U(a',g')$.
(d) If μ_m is Λ-invariant, then each $U(a,g)$ is unitary.
(e) If $g \in \mathrm{SU}(2)$, then $W(g,p) = g$.

Proof. (a) Put $q = \Lambda(g)^{-1}p$. Then h_q maps k_m to q, g maps q to p, and h_p^{-1} maps p to k_m. So $W(g,p) \in H_{k_m} = \mathrm{SU}(2)$.

(b) With q as above, $W(g,p)W(g',q) = h_p^{-1} g h_q\, h_q^{-1} g' h_{\Lambda(g')^{-1}q}$. The middle factors cancel, and $\Lambda(g')^{-1}q = \Lambda(gg')^{-1}p$.

(c) Apply (25.6) twice:

$$U(a,g)U(a',g')\psi(p) = e^{-ip\cdot a}\, e^{-iq\cdot a'}\, D_n(W(g,p))\, D_n(W(g',q))\, \psi(\Lambda(g')^{-1}q).$$

By the pairing invariance in Lemma 25.1, $q \cdot a' = p \cdot \Lambda(g)a'$, so the phase is $e^{-ip\cdot(a+\Lambda(g)a')}$. By the cocycle identity (b), the operator is $D_n(W(gg',p))$. The argument is $\Lambda(gg')^{-1}p$. By (25.2) this is $U(a+\Lambda(g)a', gg')\psi(p)$.

(d) The phase has modulus one and D_n is unitary. Therefore

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2\, d\mu_m(p) = \|\psi\|^2.$$

By the homomorphism property (c), $U(a,g)$ has inverse $U((a,g)^{-1})$.

(e) For unitary g, the matrix gh_qg^{-1} is positive definite, and its square is $g\Phi(q)g^*/m = \Phi(p)/m$. By uniqueness of the positive square root, $gh_qg^{-1} = h_p$. Hence $W(g,p) = g$. ∎

On translations, $W(I,p) = I$, so $U(a,I)$ is multiplication by $e^{-ip\cdot a}$. Hence, as in Example 25.5, the measure E of this model has $E(S)$ equal to multiplication by the indicator function of $S \cap X_m^+$. So $E(X_m^+) = I$, and $E(\{p\}) = 0$ for every point p. Formula (25.6) also has no square root of a Jacobian, unlike the induced representations of Lecture 17, because μ_m is invariant.

By the rotation case of the Wigner rotation, Proposition 25.15(e), rotations act by $(U(0,u)\psi)(p) = D_n(u)\psi(\Lambda(u)^{-1}p)$. At the rest momentum k_m the little group acts through D_n, which is spin $n/2$. The central element acts by

$$U(z) = D_n(-I) = (-1)^n I.$$

Hence half-integer spin does not descend to $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$. Time translation by t acts by multiplication by e^{-itp_0}. Its self-adjoint generator P_0 is multiplication by p_0, with domain

$$\mathrm{Dom}(P_0) = \{\psi \in \mathcal{H}_{m,n} : p_0\psi \in \mathcal{H}_{m,n}\}.$$

Since $p_0 \geq m$ on X_m^+, we get $P_0 \geq m$. The energy is positive.

Three steps complete the identification with a unitary induced representation. The measure and intertwining calculations are assigned to the sheet. Continuity is explained here.

1. *Invariance of μ_m.* Exercise 25.4 writes $g = u\, d_s\, v$ with $u, v \in \mathrm{SU}(2)$. Rotations fix p_0 and act orthogonally on $\vec{p}$. The boost d_s needs a Jacobian computation.
2. *Strong continuity.* For continuous compactly supported ψ, the formula is jointly continuous in (a,g,p). The positive square root h_p is continuous by matrix functional calculus. For (a,g) in a compact neighbourhood, transformed supports lie in one compact set and the integrands are uniformly bounded there. Dominated convergence gives norm continuity. Density and unitarity extend it to all ψ.
3. *Identification with induction.* The section map is

   $$f \longmapsto \psi_f, \qquad \psi_f(p) = f(0, h_p).$$

   It carries the induced representation from $A \rtimes \mathrm{SU}(2)$, with inducing data $\chi_{k_m} \otimes D_n$, onto (25.6). Exercise 25.5 checks the intertwining. Then the irreducibility of induced representations in Mackey's theorem, Theorem 25.10(b), shows that the model is irreducible.

Remark 25.16. The finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ admit no invariant inner product unless trivial, as proved in Lecture 24. In physics they label the components of fields. Restricted to $\mathrm{SU}(2)$, the components of a field in (j_L, j_R) carry the spins $|j_L - j_R|, \dots, j_L + j_R$ of Lecture 24. They do not classify the particle Hilbert space. For a massive particle of spin $n/2$, that space is $\mathcal{H}_{m,n}$, built from unitary $\mathrm{SU}(2)$ data on an orbit.

Self-checks

1. Is $(-1,0,0,0)$ in the orbit of k_1?
   Answer. No. $\Phi(k_1) = I$ is positive definite, but $\Phi(-1,0,0,0) = -I$ is not. Conjugation $\Phi \mapsto g\Phi g^*$ preserves positivity.
2. Do the defining matrices of H_k on $\mathbb{C}^2$ preserve some inner product?
   Answer. No. By Proposition 25.13, every finite-dimensional continuous unitary representation of H_k is trivial on N. An invariant inner product would make the defining representation such a representation. But $n(1) \neq I$.
3. In the model with $n = 2$, what are $U(z)$ and $W(u,p)$ for $u \in \mathrm{SU}(2)$?
   Answer. $U(z) = D_2(-I) = 1$, and $W(u,p) = u$ by the rotation case of the Wigner rotation, Proposition 25.15(e).

Summary

The Poincaré cover is $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$. Translations act through momentum characters $e^{-ip\cdot a}$, and $\mathrm{SL}(2,\mathbb{C})$ moves momenta by Λ. The orbits are mass shells, light cones, spacelike hyperboloids and $\{0\}$. The projection $E(S)$ selects the part of the space with momentum in S, and in the massive model it is multiplication by an indicator function. Mackey's theorem reduces the classification to little groups. It already gives the irreducible unitary representations of the $ax+b$ group and of the Heisenberg group. Up to unitary equivalence, irreducible unitary representations with E concentrated on X_m^+ are labelled by an $\mathrm{SU}(2)$ spin $n/2$ with $n \geq 0$. Finite-helicity massless representations have trivial little-group translations. Continuous-spin sectors also have positive energy and are separate. The zero-momentum orbit $\{0\}$ gives a further separate sector.

Reading

- R. Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2: the spectral-measure theorem for vector groups.
- E. P. Wigner, *On Unitary Representations of the Inhomogeneous Lorentz Group*, Annals of Mathematics 40 (1939), 149–204. Read the classification of momentum orbits and little groups.
- A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. Read the treatment of induced representations and Mackey's method for semidirect products.